

SPENVIS 4.6.14.3582 10-May-2026 09:59:26

Solar cell damage equivalent fluences

Project: DEMO1

iss_0.5sys_duration

Contents

[Summary table](#)

Mission segment 1

Trapped protons for [P_{max}](#) [V_{oc}](#) [I_{sc}](#)[Trapped electrons](#)Solar protons for [P_{max}](#) [V_{oc}](#) [I_{sc}](#)

Cell: Azur 3G30 (SR-NIEL 21eV)

Electron/proton damage ratios:

$$P_{\max} = 1134.8$$

$$V_{\text{oc}} = 2864.6$$

$$I_{\text{sc}} = 613.4$$

Coverglass

material: SiO₂density: 2.32 (g/cm³)Summary of 1 MeV equivalent electron fluences (cm⁻²)

Coverglass thickness			Total			Trapped electrons	Trapped protons			Solar protons		
g cm ⁻²	mils	micron	P _{max}	V _{oc}	I _{sc}	P _{max} , V _{oc} , I _{sc}	P _{max}	V _{oc}	I _{sc}	P _{max}	V _{oc}	I _{sc}
0.0000	0.0	0.00	4.574E+13	1.154E+14	2.473E+13	2.099E+10	4.572E+13	1.154E+14	2.471E+13	1.000E-12	1.000E-12	1.000E-12
0.0059	1.0	25.40	2.729E+11	6.600E+11	1.562E+11	1.896E+10	2.540E+11	6.411E+11	1.373E+11	1.000E-12	1.000E-12	1.000E-12
0.0177	3.0	76.20	1.130E+11	2.599E+11	6.872E+10	1.663E+10	9.637E+10	2.433E+11	5.209E+10	1.000E-12	1.000E-12	1.000E-12
0.0354	6.0	152.40	8.087E+10	1.823E+11	5.031E+10	1.435E+10	6.653E+10	1.679E+11	3.596E+10	1.000E-12	1.000E-12	1.000E-12
0.0707	12.0	304.80	6.182E+10	1.388E+11	3.860E+10	1.129E+10	5.053E+10	1.276E+11	2.731E+10	1.000E-12	1.000E-12	1.000E-12
0.1179	20.0	508.00	5.171E+10	1.173E+11	3.195E+10	8.710E+09	4.300E+10	1.085E+11	2.324E+10	1.000E-12	1.000E-12	1.000E-12
0.1768	30.0	762.00	4.556E+10	1.050E+11	2.765E+10	6.567E+09	3.900E+10	9.844E+10	2.108E+10	1.000E-12	1.000E-12	1.000E-12
0.3536	60.0	1524.00	3.641E+10	8.705E+10	2.114E+10	3.185E+09	3.322E+10	8.386E+10	1.796E+10	1.000E-12	1.000E-12	1.000E-12
User defined coverglass thickness												
0.0232	3.9	100.00	9.911E+10	2.262E+11	6.081E+10	1.569E+10	8.334E+10	2.104E+11	4.505E+10	0.000E+00	0.000E+00	0.000E+00

[Trapped protons](#) for [mission segment 1](#) for P_{max} (integration restricted to the input energy range). D(E,...) values are coarse averages introduced into the table to simplify the output information; the accuracy of the results should not be affected by this simplification.

			0.00 micron		25.40 micron		76.20 micron		152.40 micron		304.80 micron		508.00 micron		762.00 micron		1524.00 micron	
Energy Int. (MeV)	Int. Fluence (cm ⁻²)	Diff. Fluence (cm ⁻²)	D(E, 0.0)	Eq. Fluence (cm ⁻²)	D(E, 1.0)	Eq. Fluence (cm ⁻²)	D(E, 3.0)	Eq. Fluence (cm ⁻²)	D(E, 6.0)	Eq. Fluence (cm ⁻²)	D(E, 12.0)	Eq. Fluence (cm ⁻²)	D(E, 20.0)	Eq. Fluence (cm ⁻²)	D(E, 30.0)	Eq. Fluence (cm ⁻²)	D(E, 60.0)	Eq. Fluence (cm ⁻²)
0.10	0.20	3.371E+09	1.329E+09	1.82E+01	2.417E+10	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.20	0.30	2.042E+09	6.734E+08	1.22E+01	8.204E+09	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.30	0.40	1.369E+09	4.101E+08	9.17E+00	3.760E+09	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.40	0.60	9.585E+08	3.831E+08	6.82E+00	2.613E+09	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.60	0.80	5.754E+08	1.521E+08	5.02E+00	7.640E+08	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.80	1.00	4.233E+08	8.545E+07	3.99E+00	3.406E+08	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
1.00	1.20	3.378E+08	3.728E+07	3.33E+00	1.240E+08	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
1.20	1.30	3.006E+08	1.504E+07	2.95E+00	4.435E+07	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
1.30	1.40	2.855E+08	1.325E+07	2.75E+00	3.644E+07	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

1.40	1.60	2.723E+08	2.065E+07	2.51E+00	5.192E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.60	1.80	2.516E+08	1.540E+07	2.24E+00	3.456E+07	2.89E+00	4.451E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.80	2.00	2.362E+08	1.298E+07	2.03E+00	2.633E+07	2.86E+00	3.712E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.00	2.20	2.232E+08	6.172E+06	1.85E+00	1.141E+07	2.67E+00	1.647E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.20	2.40	2.171E+08	5.486E+06	1.70E+00	9.349E+06	2.46E+00	1.349E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.40	2.60	2.116E+08	4.924E+06	1.58E+00	7.802E+06	2.26E+00	1.112E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.60	2.80	2.067E+08	4.457E+06	1.48E+00	6.593E+06	2.07E+00	9.243E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.80	3.00	2.022E+08	4.063E+06	1.38E+00	5.625E+06	1.92E+00	7.797E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.00	3.20	1.981E+08	3.296E+06	1.30E+00	4.301E+06	1.78E+00	5.865E+06	1.34E+00	4.423E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.20	3.40	1.948E+08	3.046E+06	1.23E+00	3.761E+06	1.65E+00	5.025E+06	1.47E+00	4.493E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.40	3.60	1.918E+08	2.829E+06	1.17E+00	3.309E+06	1.54E+00	4.355E+06	1.50E+00	4.229E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.60	3.80	1.890E+08	2.637E+06	1.11E+00	2.927E+06	1.44E+00	3.810E+06	1.47E+00	3.890E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.80	4.00	1.863E+08	2.468E+06	1.05E+00	2.603E+06	1.36E+00	3.355E+06	1.43E+00	3.541E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.00	4.20	1.839E+08	1.659E+06	1.01E+00	1.674E+06	1.28E+00	2.131E+06	1.38E+00	2.289E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.20	4.40	1.822E+08	1.568E+06	9.69E-01	1.520E+06	1.21E+00	1.904E+06	1.32E+00	2.077E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.40	4.60	1.806E+08	1.485E+06	9.32E-01	1.384E+06	1.15E+00	1.708E+06	1.27E+00	1.894E+06	7.38E-01	1.096E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.60	4.80	1.791E+08	1.411E+06	8.97E-01	1.266E+06	1.09E+00	1.544E+06	1.22E+00	1.721E+06	8.97E-01	1.265E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.80	5.20	1.777E+08	2.665E+06	8.51E-01	2.268E+06	1.02E+00	2.731E+06	1.14E+00	3.049E+06	9.66E-01	2.574E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
5.20	5.60	1.751E+08	2.472E+06	7.97E-01	1.970E+06	9.43E-01	2.331E+06	1.05E+00	2.607E+06	9.83E-01	2.431E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
5.60	6.00	1.726E+08	2.270E+06	7.50E-01	1.702E+06	8.74E-01	1.984E+06	9.75E-01	2.213E+06	9.57E-01	2.172E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
6.00	6.40	1.703E+08	2.029E+06	7.10E-01	1.440E+06	8.15E-01	1.654E+06	9.08E-01	1.841E+06	9.16E-01	1.858E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
6.40	6.80	1.683E+08	1.884E+06	6.76E-01	1.273E+06	7.65E-01	1.441E+06	8.48E-01	1.598E+06	8.71E-01	1.641E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
6.80	7.20	1.664E+08	1.785E+06	6.45E-01	1.152E+06	7.22E-01	1.288E+06	7.97E-01	1.422E+06	8.27E-01	1.477E+06	5.75E-01	1.027E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
7.20	7.60	1.646E+08	1.698E+06	6.18E-01	1.050E+06	6.84E-01	1.162E+06	7.52E-01	1.277E+06	7.85E-01	1.334E+06	6.38E-01	1.084E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
7.60	8.00	1.629E+08	1.595E+06	5.94E-01	9.481E+05	6.52E-01	1.040E+06	7.12E-01	1.136E+06	7.46E-01	1.190E+06	6.56E-01	1.047E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
8.00	9.00	1.613E+08	3.604E+06	5.60E-01	2.017E+06	6.05E-01	2.182E+06	6.55E-01	2.359E+06	6.87E-01	2.477E+06	6.45E-01	2.324E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
9.00	10.00	1.577E+08	3.155E+06	5.18E-01	1.635E+06	5.52E-01	1.741E+06	5.89E-01	1.859E+06	6.18E-01	1.948E+06	6.10E-01	1.924E+06	4.17E-01	1.317E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00
10.00	11.00	1.546E+08	2.188E+06	4.88E-01	1.068E+06	5.12E-01	1.120E+06	5.40E-01	1.181E+06	5.64E-01	1.234E+06	5.67E-01	1.241E+06	4.85E-01	1.062E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00
11.00	12.00	1.524E+08	1.971E+06	4.65E-01	9.173E+05	4.82E-01	9.507E+05	5.02E-01	9.900E+05	5.21E-01	1.026E+06	5.28E-01	1.040E+06	4.86E-01	9.578E+05	2.48E-01	4.893E+05	0.00E+00	0.000E+00
12.00	13.00	1.504E+08	1.790E+06	4.46E-01	7.985E+05	4.59E-01	8.217E+05	4.74E-01	8.494E+05	4.87E-01	8.726E+05	4.94E-01	8.851E+05	4.70E-01	8.423E+05	3.71E-01	6.636E+05	0.00E+00	0.000E+00
13.00	14.00	1.486E+08	1.639E+06	4.29E-01	7.029E+05	4.39E-01	7.193E+05	4.51E-01	7.398E+05	4.61E-01	7.561E+05	4.66E-01	7.643E+05	4.51E-01	7.398E+05	3.92E-01	6.424E+05	0.00E+00	0.000E+00
14.00	15.00	1.470E+08	1.509E+06	4.13E-01	6.241E+05	4.22E-01	6.369E+05	4.32E-01	6.520E+05	4.40E-01	6.640E+05	4.44E-01	6.701E+05	4.33E-01	6.535E+05	3.93E-01	5.932E+05	0.00E+00	0.000E+00
15.00	16.00	1.455E+08	1.388E+06	3.99E-01	5.543E+05	4.07E-01	5.647E+05	4.15E-01	5.758E+05	4.21E-01	5.848E+05	4.25E-01	5.896E+05	4.16E-01	5.772E+05	3.87E-01	5.370E+05	0.00E+00	0.000E+00
16.00	18.00	1.441E+08	2.498E+06	3.81E-01	9.528E+05	3.87E-01	9.678E+05	3.94E-01	9.840E+05	3.99E-01	9.964E+05	4.02E-01	1.004E+06	3.95E-01	9.878E+05	3.74E-01	9.355E+05	0.00E+00	0.000E+00
18.00	20.00	1.416E+08	2.198E+06	3.60E-01	7.922E+05	3.65E-01	8.021E+05	3.70E-01	8.131E+05	3.73E-01	8.207E+05	3.76E-01	8.273E+05	3.72E-01	8.186E+05	3.58E-01	7.868E+05	2.61E-01	5.72E+05
20.00	22.00	1.394E+08	2.540E+06	3.46E-01	8.800E+05	3.49E-01	8.863E+05	3.52E-01	8.939E+05	3.54E-01	8.989E+05	3.55E-01	9.027E+05	3.53E-01	8.963E+05	3.43E-01	8.710E+05	2.80E-01	7.12E+05
22.00	24.00	1.369E+08	2.278E+06	3.38E-01	7.700E+05	3.39E-01	7.723E+05	3.40E-01	7.757E+05	3.41E-01	7.768E+05	3.40E-01	7.745E+05	3.37E-01	7.676E+05	3.29E-01	7.506E+05	2.85E-01	6.50E+05
24.00	26.00	1.346E+08	2.062E+06	3.30E-01	6.815E+05	3.31E-01	6.836E+05	3.32E-01	6.846E+05	3.32E-01	6.846E+05	3.30E-01	6.805E+05	3.25E-01	6.712E+05	3.18E-01	6.568E+05	2.84E-01	5.86E+05
26.00	28.00	1.325E+08	1.881E+06	3.24E-01	6.095E+05	3.24E-01	6.104E+05	3.24E-01	6.104E+05	3.24E-01	6.104E+05	3.22E-01	6.066E+05	3.17E-01	5.972E+05	3.10E-01	5.841E+05	2.81E-01	5.29E+05
28.00	30.00	1.306E+08	1.727E+06	3.18E-01	5.493E+05	3.18E-01	5.493E+05	3.18E-01	5.493E+05	3.18E-01	5.493E+05	3.16E-01	5.458E+05	3.11E-01	5.372E+05	3.04E-01	5.259E+05	2.78E-01	4.81E+05
30.00	34.00	1.289E+08	4.012E+06	3.10E-01	1.246E+06	3.10E-01	1.246E+06	3.10E-01	1.246E+06	3.10E-01	1.244E+06	3.08E-01	1.236E+06	3.03E-01	1.218E+06	2.97E-01	1.194E+06	2.75E-01	1.10E+06
34.00	38.00	1.249E+08	3.460E+06	3.02E-01	1.047E+06	3.02E-01	1.047E+06	3.02E-01	1.047E+06	3.01E-01	1.043E+06	2.99E-01	1.036E+06	2.95E-01	1.023E+06	2.90E-01	1.004E+06	2.72E-01	9.39E+05
38.00	42.00	1.214E+08	3.407E+06	2.95E-01	1.006E+06	2.95E-01	1.006E+06	2.95E-01	1.006E+06	2.94E-01	1.003E+06	2.92E-01	9.958E+05	2.89E-01	9.840E+05	2.84E-01	9.687E+05	2.69E-01	9.16E+05
42.00	46.00	1.180E+08	3.362E+06	2.88E-01	9.698E+05	2.88E-01	9.698E+05	2.88E-01	9.698E+05	2.88E-01	9.682E+05	2.86E-01	9.614E+05	2.84E-01	9.530E+05	2.80E-01	9.413E+05	2.67E-01	8.97E+05
46.00	50.00	1.147E+08	2.997E+06	2.82E-01	8.452E+05	2.82E-01	8.452E+05	2.82E-01	8.452E+05	2.82E-01	8.452E+05	2.80E-01	8.407E+05	2.78E-01	8.347E+05	2.75E-01	8.257E+05	2.65E-01	7.92E+05
50.00	55.00	1.117E+08	4.751E+06	2.76E-01	1.311E+06	2.76E-01	1.311E+06	2.76E-01	1.311E+06	2.76E-01	1.309E+06	2.75E-01	1.304E+06	2.73E-01	1.297E+06	2.71E-01	1.285E+		

1.00	1.20	3.378E+08	3.728E+07	3.33E+00	1.240E+08	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.20	1.30	3.006E+08	1.504E+07	2.95E+00	4.435E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.30	1.40	2.855E+08	1.325E+07	2.75E+00	3.644E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.40	1.60	2.723E+08	2.065E+07	2.51E+00	5.192E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.60	1.80	2.516E+08	1.540E+07	2.24E+00	3.456E+07	2.89E+00	4.451E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.80	2.00	2.362E+08	1.298E+07	2.03E+00	2.633E+07	2.86E+00	3.712E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.00	2.20	2.232E+08	6.172E+06	1.85E+00	1.141E+07	2.67E+00	1.647E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.20	2.40	2.171E+08	5.486E+06	1.70E+00	9.349E+06	2.46E+00	1.349E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.40	2.60	2.116E+08	4.924E+06	1.58E+00	7.802E+06	2.26E+00	1.112E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.60	2.80	2.067E+08	4.457E+06	1.48E+00	6.593E+06	2.07E+00	9.243E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.80	3.00	2.022E+08	4.063E+06	1.38E+00	5.625E+06	1.92E+00	7.797E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.00	3.20	1.981E+08	3.296E+06	1.30E+00	4.301E+06	1.78E+00	5.865E+06	1.34E+00	4.423E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.20	3.40	1.948E+08	3.046E+06	1.23E+00	3.761E+06	1.65E+00	5.025E+06	1.47E+00	4.493E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.40	3.60	1.918E+08	2.829E+06	1.17E+00	3.309E+06	1.54E+00	4.355E+06	1.50E+00	4.229E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.60	3.80	1.890E+08	2.637E+06	1.11E+00	2.927E+06	1.44E+00	3.810E+06	1.47E+00	3.890E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.80	4.00	1.863E+08	2.468E+06	1.05E+00	2.603E+06	1.36E+00	3.355E+06	1.43E+00	3.541E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.00	4.20	1.839E+08	1.659E+06	1.01E+00	1.674E+06	1.28E+00	2.131E+06	1.38E+00	2.289E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.20	4.40	1.822E+08	1.568E+06	9.69E-01	1.520E+06	1.21E+00	1.904E+06	1.32E+00	2.077E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.40	4.60	1.806E+08	1.485E+06	9.32E-01	1.384E+06	1.15E+00	1.708E+06	1.27E+00	1.894E+06	7.38E-01	1.096E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.60	4.80	1.791E+08	1.411E+06	8.97E-01	1.266E+06	1.09E+00	1.544E+06	1.22E+00	1.721E+06	8.97E-01	1.265E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.80	5.20	1.777E+08	2.665E+06	8.51E-01	2.268E+06	1.02E+00	2.731E+06	1.14E+00	3.049E+06	9.66E-01	2.574E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
5.20	5.60	1.751E+08	2.472E+06	7.97E-01	1.970E+06	9.43E-01	2.331E+06	1.05E+00	2.607E+06	9.83E-01	2.431E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
5.60	6.00	1.726E+08	2.270E+06	7.50E-01	1.702E+06	8.74E-01	1.984E+06	9.75E-01	2.213E+06	9.57E-01	2.172E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
6.00	6.40	1.703E+08	2.029E+06	7.10E-01	1.440E+06	8.15E-01	1.654E+06	9.08E-01	1.841E+06	9.16E-01	1.858E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
6.40	6.80	1.683E+08	1.884E+06	6.76E-01	1.273E+06	7.65E-01	1.441E+06	8.48E-01	1.598E+06	8.71E-01	1.641E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
6.80	7.20	1.664E+08	1.785E+06	6.45E-01	1.152E+06	7.22E-01	1.288E+06	7.97E-01	1.422E+06	8.27E-01	1.477E+06	5.75E-01	1.027E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
7.20	7.60	1.646E+08	1.698E+06	6.18E-01	1.050E+06	6.84E-01	1.162E+06	7.52E-01	1.277E+06	7.85E-01	1.334E+06	6.38E-01	1.084E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
7.60	8.00	1.629E+08	1.595E+06	5.94E-01	9.481E+05	6.52E-01	1.040E+06	7.12E-01	1.136E+06	7.46E-01	1.190E+06	6.56E-01	1.047E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
8.00	9.00	1.613E+08	3.604E+06	5.60E-01	2.017E+06	6.05E-01	2.182E+06	6.55E-01	2.359E+06	6.87E-01	2.477E+06	6.45E-01	2.324E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
9.00	10.00	1.577E+08	3.155E+06	5.18E-01	1.635E+06	5.52E-01	1.741E+06	5.89E-01	1.859E+06	6.18E-01	1.948E+06	6.10E-01	1.924E+06	4.17E-01	1.317E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00
10.00	11.00	1.546E+08	2.188E+06	4.88E-01	1.068E+06	5.12E-01	1.120E+06	5.40E-01	1.181E+06	5.64E-01	1.234E+06	5.67E-01	1.241E+06	4.85E-01	1.062E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00
11.00	12.00	1.524E+08	1.971E+06	4.65E-01	9.173E+05	4.82E-01	9.507E+05	5.02E-01	9.900E+05	5.21E-01	1.026E+06	5.28E-01	1.040E+06	4.86E-01	9.578E+05	2.48E-01	4.893E+05	0.00E+00	0.000E+00
12.00	13.00	1.504E+08	1.790E+06	4.46E-01	7.985E+05	4.59E-01	8.217E+05	4.74E-01	8.494E+05	4.87E-01	8.726E+05	4.94E-01	8.851E+05	4.70E-01	8.423E+05	3.71E-01	6.636E+05	0.00E+00	0.000E+00
13.00	14.00	1.486E+08	1.639E+06	4.29E-01	7.029E+05	4.39E-01	7.193E+05	4.51E-01	7.398E+05	4.61E-01	7.561E+05	4.66E-01	7.643E+05	4.51E-01	7.398E+05	3.92E-01	6.424E+05	0.00E+00	0.000E+00
14.00	15.00	1.470E+08	1.509E+06	4.13E-01	6.241E+05	4.22E-01	6.369E+05	4.32E-01	6.520E+05	4.40E-01	6.640E+05	4.44E-01	6.701E+05	4.33E-01	6.535E+05	3.93E-01	5.932E+05	0.00E+00	0.000E+00
15.00	16.00	1.455E+08	1.388E+06	3.99E-01	5.543E+05	4.07E-01	5.647E+05	4.15E-01	5.758E+05	4.21E-01	5.848E+05	4.25E-01	5.896E+05	4.16E-01	5.772E+05	3.87E-01	5.370E+05	0.00E+00	0.000E+00
16.00	18.00	1.441E+08	2.498E+06	3.81E-01	9.528E+05	3.87E-01	9.678E+05	3.94E-01	9.840E+05	3.99E-01	9.964E+05	4.02E-01	1.004E+06	3.95E-01	9.878E+05	3.74E-01	9.355E+05	0.00E+00	0.000E+00
18.00	20.00	1.416E+08	2.198E+06	3.60E-01	7.922E+05	3.65E-01	8.021E+05	3.70E-01	8.131E+05	3.73E-01	8.207E+05	3.76E-01	8.273E+05	3.72E-01	8.186E+05	3.58E-01	7.868E+05	2.61E-01	5.72E+05
20.00	22.00	1.394E+08	2.540E+06	3.46E-01	8.800E+05	3.49E-01	8.863E+05	3.52E-01	8.939E+05	3.54E-01	8.989E+05	3.55E-01	9.027E+05	3.53E-01	8.963E+05	3.43E-01	8.710E+05	2.80E-01	7.12E+05
22.00	24.00	1.369E+08	2.278E+06	3.38E-01	7.700E+05	3.39E-01	7.723E+05	3.40E-01	7.757E+05	3.41E-01	7.768E+05	3.40E-01	7.745E+05	3.37E-01	7.676E+05	3.29E-01	7.506E+05	2.85E-01	6.50E+05
24.00	26.00	1.346E+08	2.062E+06	3.30E-01	6.815E+05	3.31E-01	6.836E+05	3.32E-01	6.846E+05	3.32E-01	6.846E+05	3.30E-01	6.805E+05	3.25E-01	6.712E+05	3.18E-01	6.568E+05	2.84E-01	5.86E+05
26.00	28.00	1.325E+08	1.881E+06	3.24E-01	6.095E+05	3.24E-01	6.104E+05	3.24E-01	6.104E+05	3.24E-01	6.104E+05	3.22E-01	6.066E+05	3.17E-01	5.972E+05	3.10E-01	5.841E+05	2.81E-01	5.29E+05
28.00	30.00	1.306E+08	1.727E+06	3.18E-01	5.493E+05	3.18E-01	5.493E+05	3.18E-01	5.493E+05	3.18E-01	5.493E+05	3.16E-01	5.458E+05	3.11E-01	5.372E+05	3.04E-01	5.259E+05	2.78E-01	4.81E+05
30.00	34.00	1.289E+08	4.012E+06	3.10E-01	1.246E+06	3.10E-01	1.246E+06	3.10E-01	1.246E+06	3.10E-01	1.244E+06	3.08E-01	1.236E+06	3.03E-01	1.218E+06	2.97E-01	1.194E+06	2.75E-01	1.10E+06
34.00	38.00	1.249E+08	3.460E+06	3.02E-01	1.047E+06	3.02E-01	1.047E+06	3.02E-01	1.047E+06	3.01E-01	1.043E+06	2.99E-01	1.036E+06	2.95E-01	1.023E+06	2.90E-01	1.004E+06	2.72E-01	9.39E+05
38.00	42.00	1.214E+08	3.407E+06	2.95E-01	1.006E+06	2.95E-01	1.006E+06	2.95E-01	1.006E+06	2.94E-01	1.003E+06	2.92E-01	9.958E+05	2.89E-01	9.840E+05	2.84E-01	9.687E+05</		

0.60	0.80	5.754E+08	1.521E+08	5.02E+00	7.640E+08	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
0.80	1.00	4.233E+08	8.545E+07	3.99E+00	3.406E+08	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.00	1.20	3.378E+08	3.728E+07	3.33E+00	1.240E+08	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.20	1.30	3.006E+08	1.504E+07	2.95E+00	4.435E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.30	1.40	2.855E+08	1.325E+07	2.75E+00	3.644E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.40	1.60	2.723E+08	2.065E+07	2.51E+00	5.192E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.60	1.80	2.516E+08	1.540E+07	2.24E+00	3.456E+07	2.89E+00	4.451E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.80	2.00	2.362E+08	1.298E+07	2.03E+00	2.633E+07	2.86E+00	3.712E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.00	2.20	2.232E+08	6.172E+06	1.85E+00	1.141E+07	2.67E+00	1.647E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.20	2.40	2.171E+08	5.486E+06	1.70E+00	9.349E+06	2.46E+00	1.349E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.40	2.60	2.116E+08	4.924E+06	1.58E+00	7.802E+06	2.26E+00	1.112E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.60	2.80	2.067E+08	4.457E+06	1.48E+00	6.593E+06	2.07E+00	9.243E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.80	3.00	2.022E+08	4.063E+06	1.38E+00	5.625E+06	1.92E+00	7.797E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.00	3.20	1.981E+08	3.296E+06	1.30E+00	4.301E+06	1.78E+00	5.865E+06	1.34E+00	4.423E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.20	3.40	1.948E+08	3.046E+06	1.23E+00	3.761E+06	1.65E+00	5.025E+06	1.47E+00	4.493E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.40	3.60	1.918E+08	2.829E+06	1.17E+00	3.309E+06	1.54E+00	4.355E+06	1.50E+00	4.229E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.60	3.80	1.890E+08	2.637E+06	1.11E+00	2.927E+06	1.44E+00	3.810E+06	1.47E+00	3.890E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.80	4.00	1.863E+08	2.468E+06	1.05E+00	2.603E+06	1.36E+00	3.355E+06	1.43E+00	3.541E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.00	4.20	1.839E+08	1.659E+06	1.01E+00	1.674E+06	1.28E+00	2.131E+06	1.38E+00	2.289E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.20	4.40	1.822E+08	1.568E+06	9.69E-01	1.520E+06	1.21E+00	1.904E+06	1.32E+00	2.077E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.40	4.60	1.806E+08	1.485E+06	9.32E-01	1.384E+06	1.15E+00	1.708E+06	1.27E+00	1.894E+06	7.38E-01	1.096E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.60	4.80	1.791E+08	1.411E+06	8.97E-01	1.266E+06	1.09E+00	1.544E+06	1.22E+00	1.721E+06	8.97E-01	1.265E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.80	5.20	1.777E+08	2.665E+06	8.51E-01	2.268E+06	1.02E+00	2.731E+06	1.14E+00	3.049E+06	9.66E-01	2.574E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
5.20	5.60	1.751E+08	2.472E+06	7.97E-01	1.970E+06	9.43E-01	2.331E+06	1.05E+00	2.607E+06	9.83E-01	2.431E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
5.60	6.00	1.726E+08	2.270E+06	7.50E-01	1.702E+06	8.74E-01	1.984E+06	9.75E-01	2.213E+06	9.57E-01	2.172E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
6.00	6.40	1.703E+08	2.029E+06	7.10E-01	1.440E+06	8.15E-01	1.654E+06	9.08E-01	1.841E+06	9.16E-01	1.858E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
6.40	6.80	1.683E+08	1.884E+06	6.76E-01	1.273E+06	7.65E-01	1.441E+06	8.48E-01	1.598E+06	8.71E-01	1.641E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
6.80	7.20	1.664E+08	1.785E+06	6.45E-01	1.152E+06	7.22E-01	1.288E+06	7.97E-01	1.422E+06	8.27E-01	1.477E+06	5.75E-01	1.027E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
7.20	7.60	1.646E+08	1.698E+06	6.18E-01	1.050E+06	6.84E-01	1.162E+06	7.52E-01	1.277E+06	7.85E-01	1.334E+06	6.38E-01	1.084E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
7.60	8.00	1.629E+08	1.595E+06	5.94E-01	9.481E+05	6.52E-01	1.040E+06	7.12E-01	1.136E+06	7.46E-01	1.190E+06	6.56E-01	1.047E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
8.00	9.00	1.613E+08	3.604E+06	5.60E-01	2.017E+06	6.05E-01	2.182E+06	6.55E-01	2.359E+06	6.87E-01	2.477E+06	6.45E-01	2.324E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
9.00	10.00	1.577E+08	3.155E+06	5.18E-01	1.635E+06	5.52E-01	1.741E+06	5.89E-01	1.859E+06	6.18E-01	1.948E+06	6.10E-01	1.924E+06	4.17E-01	1.317E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00
10.00	11.00	1.546E+08	2.188E+06	4.88E-01	1.068E+06	5.12E-01	1.120E+06	5.40E-01	1.181E+06	5.64E-01	1.234E+06	5.67E-01	1.241E+06	4.85E-01	1.062E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00
11.00	12.00	1.524E+08	1.971E+06	4.65E-01	9.173E+05	4.82E-01	9.507E+05	5.02E-01	9.900E+05	5.21E-01	1.026E+06	5.28E-01	1.040E+06	4.86E-01	9.578E+05	2.48E-01	4.893E+05	0.00E+00	0.000E+00
12.00	13.00	1.504E+08	1.790E+06	4.46E-01	7.985E+05	4.59E-01	8.217E+05	4.74E-01	8.494E+05	4.87E-01	8.726E+05	4.94E-01	8.851E+05	4.70E-01	8.423E+05	3.71E-01	6.636E+05	0.00E+00	0.000E+00
13.00	14.00	1.486E+08	1.639E+06	4.29E-01	7.029E+05	4.39E-01	7.193E+05	4.51E-01	7.398E+05	4.61E-01	7.561E+05	4.66E-01	7.643E+05	4.51E-01	7.398E+05	3.92E-01	6.424E+05	0.00E+00	0.000E+00
14.00	15.00	1.470E+08	1.509E+06	4.13E-01	6.241E+05	4.22E-01	6.369E+05	4.32E-01	6.520E+05	4.40E-01	6.640E+05	4.44E-01	6.701E+05	4.33E-01	6.535E+05	3.93E-01	5.932E+05	0.00E+00	0.000E+00
15.00	16.00	1.455E+08	1.388E+06	3.99E-01	5.543E+05	4.07E-01	5.647E+05	4.15E-01	5.758E+05	4.21E-01	5.848E+05	4.25E-01	5.896E+05	4.16E-01	5.772E+05	3.87E-01	5.370E+05	0.00E+00	0.000E+00
16.00	18.00	1.441E+08	2.498E+06	3.81E-01	9.528E+05	3.87E-01	9.678E+05	3.94E-01	9.840E+05	3.99E-01	9.964E+05	4.02E-01	1.004E+06	3.95E-01	9.878E+05	3.74E-01	9.355E+05	0.00E+00	0.000E+00
18.00	20.00	1.416E+08	2.198E+06	3.60E-01	7.922E+05	3.65E-01	8.021E+05	3.70E-01	8.131E+05	3.73E-01	8.207E+05	3.76E-01	8.273E+05	3.72E-01	8.186E+05	3.58E-01	7.868E+05	2.61E-01	5.72E+05
20.00	22.00	1.394E+08	2.540E+06	3.46E-01	8.800E+05	3.49E-01	8.863E+05	3.52E-01	8.939E+05	3.54E-01	8.989E+05	3.55E-01	9.027E+05	3.53E-01	8.963E+05	3.43E-01	8.710E+05	2.80E-01	7.12E+05
22.00	24.00	1.369E+08	2.278E+06	3.38E-01	7.700E+05	3.39E-01	7.723E+05	3.40E-01	7.757E+05	3.41E-01	7.768E+05	3.40E-01	7.745E+05	3.37E-01	7.676E+05	3.29E-01	7.506E+05	2.85E-01	6.50E+05
24.00	26.00	1.346E+08	2.062E+06	3.30E-01	6.815E+05	3.31E-01	6.836E+05	3.32E-01	6.846E+05	3.32E-01	6.846E+05	3.30E-01	6.805E+05	3.25E-01	6.712E+05	3.18E-01	6.568E+05	2.84E-01	5.86E+05
26.00	28.00	1.325E+08	1.881E+06	3.24E-01	6.095E+05	3.24E-01	6.104E+05	3.24E-01	6.104E+05	3.24E-01	6.104E+05	3.22E-01	6.066E+05	3.17E-01	5.972E+05	3.10E-01	5.841E+05	2.81E-01	5.29E+05
28.00	30.00	1.306E+08	1.727E+06	3.18E-01	5.493E+05	3.18E-01	5.493E+05	3.18E-01	5.493E+05	3.18E-01	5.493E+05	3.16E-01	5.458E+05	3.11E-01	5.372E+05	3.04E-01	5.259E+05	2.78E-01	4.81E+05
30.00	34.00	1.289E+08	4.012E+06	3.10E-01	1.246E+06	3.10E-01	1.246E+06	3.10E-01	1.246E+06	3.10E-01	1.244E+06	3.08E-01	1.236E+06	3.03E-01	1.218E+06	2.97E-01	1.194E+06		

0.17	0.18	4.888E+11	4.559E+10	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.18	0.19	4.433E+11	4.133E+10	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.19	0.20	4.019E+11	3.748E+10	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.20	0.22	3.644E+11	5.453E+10	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.22	0.24	3.099E+11	4.637E+10	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.24	0.26	2.635E+11	3.943E+10	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.26	0.28	2.241E+11	3.353E+10	6.12E-05	2.052E+06	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.28	0.30	1.906E+11	2.852E+10	2.00E-04	5.715E+06	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.30	0.32	1.621E+11	1.896E+10	3.60E-04	6.818E+06	1.56E-04	2.954E+06	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.32	0.36	1.431E+11	3.153E+10	8.10E-04	2.553E+07	3.77E-04	1.189E+07	8.11E-05	2.557E+06	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.36	0.40	1.116E+11	2.458E+10	2.13E-03	5.232E+07	1.08E-03	2.646E+07	4.10E-04	1.008E+07	6.86E-05	1.685E+06	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.40	0.45	8.697E+10	1.989E+10	5.67E-03	1.128E+08	3.08E-03	6.133E+07	1.31E-03	2.609E+07	4.09E-04	8.141E+06	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.45	0.50	6.709E+10	1.534E+10	1.49E-02	2.289E+08	8.64E-03	1.326E+08	4.05E-03	6.220E+07	1.47E-03	2.247E+07	1.83E-04	2.811E+06	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.50	0.60	5.175E+10	1.388E+10	5.49E-02	7.619E+08	3.43E-02	4.760E+08	1.79E-02	2.490E+08	7.65E-03	1.062E+08	1.44E-03	2.005E+07	9.72E-05	1.350E+06	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.60	0.70	3.786E+10	9.308E+09	1.57E-01	1.457E+09	1.14E-01	1.058E+09	7.26E-02	6.758E+08	4.00E-02	3.720E+08	1.12E-02	1.046E+08	1.67E-03	1.552E+07	7.60E-05	7.071E+05	0.00E+00	0.00E+00
0.70	0.80	2.856E+10	5.924E+09	2.56E-01	1.514E+09	2.13E-01	1.262E+09	1.62E-01	9.622E+08	1.11E-01	6.597E+08	4.73E-02	2.801E+08	1.21E-02	7.194E+07	1.34E-03	7.961E+06	0.00E+00	0.00E+00
0.80	0.90	2.263E+10	3.975E+09	3.55E-01	1.410E+09	3.11E-01	1.235E+09	2.54E-01	1.011E+09	1.95E-01	7.736E+08	1.12E-01	4.444E+08	4.65E-02	1.848E+08	1.01E-02	4.000E+07	0.00E+00	0.00E+00
0.90	1.00	1.866E+10	3.277E+09	4.52E-01	1.482E+09	4.07E-01	1.334E+09	3.47E-01	1.138E+09	2.81E-01	9.209E+08	1.85E-01	6.070E+08	1.02E-01	3.332E+08	3.83E-02	1.255E+08	7.32E-05	2.400E+00
1.00	1.20	1.538E+10	4.065E+09	5.61E-01	2.280E+09	5.20E-01	2.114E+09	4.63E-01	1.883E+09	3.96E-01	1.612E+09	2.94E-01	1.193E+09	1.94E-01	7.867E+08	1.04E-01	4.242E+08	3.57E-03	1.452E+00
1.20	1.40	1.131E+10	2.972E+09	6.93E-01	2.059E+09	6.56E-01	1.950E+09	6.03E-01	1.793E+09	5.40E-01	1.604E+09	4.37E-01	1.299E+09	3.29E-01	9.785E+08	2.23E-01	6.616E+08	3.91E-02	1.161E+00
1.40	1.60	8.343E+09	2.229E+09	8.31E-01	1.853E+09	7.93E-01	1.768E+09	7.38E-01	1.646E+09	6.72E-01	1.498E+09	5.66E-01	1.262E+09	4.54E-01	1.013E+09	3.43E-01	7.655E+08	1.17E-01	2.598E+00
1.60	1.80	6.114E+09	1.663E+09	9.74E-01	1.619E+09	9.34E-01	1.552E+09	8.76E-01	1.456E+09	8.06E-01	1.339E+09	6.94E-01	1.153E+09	5.75E-01	9.561E+08	4.56E-01	7.589E+08	2.08E-01	3.459E+00
1.80	2.00	4.451E+09	1.207E+09	1.12E+00	1.353E+09	1.08E+00	1.304E+09	1.02E+00	1.226E+09	9.45E-01	1.140E+09	8.26E-01	9.968E+08	7.00E-01	8.443E+08	5.72E-01	6.904E+08	3.02E-01	3.650E+00
2.00	2.25	3.244E+09	1.030E+09	1.26E+00	1.295E+09	1.22E+00	1.258E+09	1.16E+00	1.191E+09	1.09E+00	1.123E+09	9.69E-01	9.984E+08	8.40E-01	8.654E+08	7.04E-01	7.252E+08	4.09E-01	4.210E+00
2.25	2.50	2.214E+09	7.004E+08	1.38E+00	9.644E+08	1.35E+00	9.433E+08	1.29E+00	9.045E+08	1.23E+00	8.592E+08	1.11E+00	7.783E+08	9.88E-01	6.919E+08	8.50E-01	5.952E+08	5.33E-01	3.732E+00
2.50	2.75	1.514E+09	5.546E+08	1.49E+00	8.274E+08	1.46E+00	8.108E+08	1.41E+00	7.830E+08	1.35E+00	7.467E+08	1.24E+00	6.857E+08	1.12E+00	6.190E+08	9.83E-01	5.453E+08	6.61E-01	3.668E+00
2.75	3.00	9.593E+08	3.453E+08	1.61E+00	5.547E+08	1.58E+00	5.443E+08	1.53E+00	5.270E+08	1.47E+00	5.063E+08	1.36E+00	4.683E+08	1.24E+00	4.268E+08	1.11E+00	3.819E+08	7.86E-01	2.714E+00
3.00	3.25	6.140E+08	2.254E+08	1.71E+00	3.849E+08	1.68E+00	3.792E+08	1.63E+00	3.679E+08	1.58E+00	3.554E+08	1.47E+00	3.317E+08	1.35E+00	3.046E+08	1.22E+00	2.753E+08	9.01E-01	2.030E+00
3.25	3.50	3.886E+08	1.411E+08	1.79E+00	2.530E+08	1.77E+00	2.495E+08	1.72E+00	2.431E+08	1.67E+00	2.360E+08	1.57E+00	2.219E+08	1.46E+00	2.056E+08	1.33E+00	1.873E+08	1.01E+00	1.420E+00
3.50	3.75	2.475E+08	9.930E+07	1.88E+00	1.864E+08	1.85E+00	1.835E+08	1.81E+00	1.795E+08	1.76E+00	1.745E+08	1.66E+00	1.650E+08	1.55E+00	1.541E+08	1.43E+00	1.417E+08	1.11E+00	1.103E+00
3.75	4.00	1.482E+08	5.894E+07	1.96E+00	1.154E+08	1.93E+00	1.139E+08	1.89E+00	1.115E+08	1.84E+00	1.083E+08	1.75E+00	1.030E+08	1.64E+00	9.652E+07	1.52E+00	8.944E+07	1.21E+00	7.143E+00
4.00	4.50	8.928E+07	6.100E+07	2.06E+00	1.254E+08	2.04E+00	1.242E+08	2.00E+00	1.220E+08	1.94E+00	1.186E+08	1.85E+00	1.131E+08	1.75E+00	1.067E+08	1.63E+00	9.959E+07	1.33E+00	8.126E+00
4.50	5.00	2.828E+07	1.964E+07	2.19E+00	4.293E+07	2.17E+00	4.253E+07	2.14E+00	4.194E+07	2.09E+00	4.096E+07	2.00E+00	3.928E+07	1.90E+00	3.731E+07	1.79E+00	3.515E+07	1.50E+00	2.942E+00
5.00	5.50	8.644E+06	6.721E+06	2.31E+00	1.549E+07	2.29E+00	1.536E+07	2.26E+00	1.516E+07	2.21E+00	1.485E+07	2.13E+00	1.431E+07	2.03E+00	1.367E+07	1.92E+00	1.293E+07	1.65E+00	1.107E+00
5.50	6.00	1.923E+06	1.923E+06	2.02E+00	3.886E+06	2.00E+00	3.854E+06	1.98E+00	3.806E+06	1.94E+00	3.735E+06	1.88E+00	3.612E+06	1.80E+00	3.467E+06	1.71E+00	3.296E+06	1.49E+00	2.868E+00
6.00	7.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00
7.00	8.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00
8.00	9.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00
9.00	10.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00
10.00	15.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00
15.00	20.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00
20.00	25.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00
25.00	30.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00
30.00	40.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00
Equiv. 1 MeV Electron Fluence (cm ⁻²)					2.099E+10		1.896E+10		1.663E+10		1.435E+10		1.129E+10		8.710E+09		6.567E+09		3.185E+00

Solar protons for P_{max} (integration restricted to the input energy range). D(E,...) values are coarse averages introduced into the table to simplify the output information; the accuracy of the results should not be affected by this simplification.

				0.00 micron		25.40 micron		76.20 micron		152.40 micron		304.80 micron		508.00 micron		762.00 micron		1524.00 micron	
Energy Int. (MeV)	Int. Fluence (cm ⁻²)	Diff. Fluence (cm ⁻²)	D(E, 0.0)	Eq. Fluence (cm ⁻²)	D(E, 1.0)	Eq. Fluence (cm ⁻²)	D(E, 3.0)	Eq. Fluence (cm ⁻²)	D(E, 6.0)	Eq. Fluence (cm ⁻²)	D(E, 12.0)	Eq. Fluence (cm ⁻²)	D(E, 20.0)	Eq. Fluence (cm ⁻²)	D(E, 30.0)	Eq. Fluence (cm ⁻²)	D(E, 60.0)	Eq. Fluence (cm ⁻²)	
0.10	0.20	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.20	0.30	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.30	0.40	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.40	0.60	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.60	0.80	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.80	1.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
1.00	1.20	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
1.20	1.30	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
1.30	1.40	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
1.40	1.60	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
1.60	1.80	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
1.80	2.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
2.00	2.20	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
2.20	2.40	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
2.40	2.60	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
2.60	2.80	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
2.80	3.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
3.00	3.20	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
3.20	3.40	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

Solar protons for V_{oc} (integration restricted to the input energy range). D(E,..) values are coarse averages introduced into the table to simplify the output information; the accuracy of the results should not be affected by this simplification.

			0.00 micron		25.40 micron		76.20 micron		152.40 micron		304.80 micron		508.00 micron		762.00 micron		1524.00 micron	
Energy Int. (MeV)	Int. Fluence (cm ⁻²)	Diff. Fluence (cm ⁻²)	D(E, 0.0)	Eq. Fluence (cm ⁻²)	D(E, 1.0)	Eq. Fluence (cm ⁻²)	D(E, 3.0)	Eq. Fluence (cm ⁻²)	D(E, 6.0)	Eq. Fluence (cm ⁻²)	D(E,12.0)	Eq. Fluence (cm ⁻²)	D(E,20.0)	Eq. Fluence (cm ⁻²)	D(E,30.0)	Eq. Fluence (cm ⁻²)	D(E,60.0)	Eq. Fluence (cm ⁻²)
0.10	0.20	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
0.20	0.30	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
0.30	0.40	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
0.40	0.60	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
0.60	0.80	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
0.80	1.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.00	1.20	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.20	1.30	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.30	1.40	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.40	1.60	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.60	1.80	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.80	2.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.00	2.20	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.20	2.40	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.40	2.60	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.60	2.80	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00

simplify the output information; the accuracy of the results should not be affected by this simplification.

[illegible]

Solar protons for I_{sc} (integration restricted to the input energy range). $D(E,...)$ values are coarse averages introduced into the table to simplify the output information; the accuracy of the results should not be affected by this simplification.

https://www.spenvis.oma.be/spenvis/htusers/k/kawakita/DEMO1/1778399966/spenvis_efp.html 7/8

